

Jaypee Institute of Information Technology, Noida



Project Title: Statistical Analysis of Factors Affecting Student Exam Performance

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1. ABSTRACT

Academic performance of students is influenced by multiple behavioral, environmental, and personal factors. This study analyzes the factors affecting student academic performance using data analysis and machine learning techniques. The dataset contains various attributes such as age, study hours, social media usage, attendance percentage, sleep hours, diet quality, parental education level, and exam scores.

Exploratory Data Analysis (EDA) techniques including histograms, scatter plots, and correlation heatmaps were applied to understand data distribution and relationships between variables. Data preprocessing steps such as handling missing values, encoding categorical variables, and feature scaling were performed to prepare the dataset for modeling.

Multiple machine learning regression models including Linear Regression, Decision Tree, Random Forest, AdaBoost, and XGBoost were implemented to predict student exam scores. The performance of these models was evaluated using metrics such as R^2 score and Root Mean Squared Error (RMSE).

The results indicate that study hours and attendance positively influence exam performance, while excessive social media usage negatively impacts academic outcomes. Among the models, Linear Regression and XGBoost demonstrated strong predictive performance. This study highlights the importance of lifestyle and behavioral factors in determining academic success and demonstrates the effectiveness of machine learning models in predicting student performance.

Keywords: Exploratory Data Analysis, Machine Learning, Student Performance, Statistical Analysis, Academic Performance.

2. INTRODUCTION

Education plays a crucial role in the development of individuals and society. Academic performance is one of the most important indicators of student success in educational institutions. Understanding the factors that influence student performance can help educators, policymakers, and students improve learning strategies and outcomes.

Student performance is affected by various factors such as study habits, lifestyle choices, attendance, and access to educational resources. With the advancement of data science and machine learning, it has become possible to analyze educational data and build predictive models that estimate student performance based on these factors.

This project focuses on analyzing a dataset that includes information about students' daily habits, lifestyle factors, and academic scores. The goal of this study is to identify important features influencing academic performance and to develop machine learning models capable of predicting exam scores.

The process begins with data preprocessing, including handling missing values, encoding categorical variables, and feature scaling. Exploratory Data Analysis (EDA) is then performed to understand the distribution of variables and relationships among them. Various regression models such as Linear Regression, Decision Tree, Random Forest, AdaBoost, and XGBoost are applied to predict student performance.

The performance of these models is evaluated using standard evaluation metrics such as R^2 score and RMSE. By comparing multiple models, the study identifies the most effective approach for predicting student scores.

The findings from this study provide valuable insights into the key factors influencing academic performance and demonstrate how machine learning techniques can be used to build predictive systems that support educational decision-making.

3. LITERATURE SURVEY

1. Prediction of Student's Performance With Learning Coefficients Using Regression Based Machine Learning Models [1]

This paper uses regression-based machine learning models such as Linear Regression, Decision Tree, Random Forest, and Support Vector Regression (SVR) to predict student performance. The factors considered include academic data (CGPA, semester grades, 10th & 12th scores), demographic data (gender), and a newly proposed feature called learning coefficients, which are calculated through adaptive assessments during the semester to measure students' learning progress. The result shows that Linear Regression performed the best with about 97% accuracy. The study concludes that adding learning coefficients improves prediction accuracy and also helps in identifying areas where students can improve.

2. Leveraging Machine and Ensemble Learning Techniques to Timely Predict Student Academic Achievements and Performance [2]

This paper uses various machine learning and ensemble learning models such as Support Vector Machine (SVM), Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Naive Bayes, Logistic Regression, Multi-Layer Perceptron (MLP), AdaBoost, and Ensemble (Voting Classifier) to predict student performance. The factors considered include academic data (CGPA, SGPA, 10th & 12th marks), behavioral data (student behavior, study hours, social media usage), lifestyle factors (smoking, alcohol, extracurricular activities), and family background (parents' education, income, etc.). The results show that SVM, Random Forest, Decision Tree, MLP, and Ensemble models achieved 100% accuracy, while Naive Bayes and AdaBoost performed the worst. The paper concludes that ensemble and advanced ML models provide highly accurate predictions and can help improve teaching strategies and student performance.

3. An Ensemble Prediction Model for Potential Student Recommendation Using Machine Learning [3]

This paper proposes an ensemble machine learning model to predict student performance and recommend suitable actions for improvement. The models used include Decision Tree, Random Forest, Logistic Regression, and ensemble techniques that combine multiple models for better accuracy. The factors considered include academic performance (marks, grades), student behavior, attendance, and other learning-related attributes that influence performance. The result shows that the ensemble model outperforms individual models in prediction accuracy. The paper concludes that combining multiple machine learning models gives more reliable predictions and can be effectively used for student recommendation systems and early intervention.

4. A Study on Regression Based Machine Learning Models to Predict the Student Performance [4]

This paper analyzes different regression-based machine learning models such as Linear Regression, Decision Tree Regression, and Random Forest Regression to predict student performance. The factors considered include academic details (marks, grades, attendance), along with student-related attributes like study time and other influencing parameters. The result shows that Random Forest and Decision Tree performed better than basic Linear Regression in terms of prediction accuracy. The study concludes that regression models are effective for predicting student performance, and choosing the right model can significantly improve accuracy.

5. A Comparative Analysis of Machine Learning Models for Student Performance Prediction [5]

This paper compares different machine learning models such as Linear Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) for predicting student performance. The factors considered include academic data (marks, attendance), behavioral factors (study time, participation), and demographic details. The result shows that ensemble models like Random Forest performed better than individual models, providing higher accuracy. The study concludes that choosing the right model and combining multiple models improves prediction performance.

4. RESEARCH GAP

Although many studies have analyzed student academic performance, several gaps still exist:

- Many studies focus primarily on traditional factors such as socioeconomic background, while giving less attention to lifestyle behaviors like social media usage, sleep patterns, and daily study habits.
- Some research applies machine learning models without performing sufficient exploratory data analysis to understand data distributions and relationships between variables.
- Several studies rely on a single predictive model rather than comparing multiple machine learning algorithms to identify the most effective approach.
- Limited work has been done on integrating data preprocessing techniques, feature engineering, and model evaluation within a single structured framework.
- Few studies provide practical implementation, such as user-interactive systems, to demonstrate real-time prediction of student performance.

This study addresses these gaps by combining exploratory data analysis with multiple machine learning regression models and developing a predictive system that can estimate student exam scores based on input features.

5.NOVELTY OF THE STUDY

The novelty of this work lies in the integration of data analysis and machine learning techniques to evaluate and predict student academic performance.

Key contributions include:

- Systematic preprocessing and transformation of dataset variables for machine learning models.
- Comprehensive exploratory data analysis using multiple visualization techniques to understand data patterns.
- Implementation and comparison of multiple regression models within a single study, including Linear Regression, Decision Tree, Random Forest, AdaBoost, and XGBoost.
- Identification of key behavioral and lifestyle factors influencing academic performance through feature analysis.
- Development of a predictive system capable of estimating student exam scores based on input attributes.

This combined approach provides both analytical insights and predictive capabilities, offering a practical framework for understanding and forecasting student performance.

6. OBJECTIVES

The main objectives of this study are:

1. To preprocess and prepare the dataset using appropriate encoding and scaling techniques.
2. To perform exploratory data analysis to understand data distributions and relationships among variables.
3. To identify key lifestyle and behavioral factors affecting academic performance.
4. To develop and implement multiple machine learning regression models for predicting exam scores.
5. To evaluate and compare model performance using metrics such as R^2 score and RMSE.
6. To build a predictive system that can estimate student performance based on user input.

7. METHODOLOGY

The methodology followed in this study consists of several stages.

1. Dataset Description

The dataset used in this project is titled Student Habits vs Academic Performance. It contains information about student demographics, lifestyle habits, and exam scores. The dataset includes variables such as age, gender, study hours per day, social media usage, sleep hours, attendance percentage, diet quality, and participation in extracurricular activities.

The dataset contains approximately 1000 observations and 16 attributes, including both categorical and numerical variables.

2. Data Collection

The dataset contains information about student characteristics and lifestyle factors including:

- Age
- Study hours per day
- Social media hours
- Netflix hours
- Attendance percentage
- Sleep hours
- Diet quality
- Exercise frequency
- Parental education level
- Internet quality
- Mental health rating
- Extracurricular participation
- Exam score

3. Attribute Classification

To better understand the dataset and select appropriate statistical techniques, each attribute was classified based on its data type, subtype, and measurement scale. Attributes are broadly categorized as qualitative (categorical) or quantitative (numerical). Qualitative variables describe categories or labels, while quantitative variables represent measurable numerical values. The classification also identifies whether variables follow nominal, ordinal, interval, or ratio scales, which helps determine suitable statistical analyses.

Table 1: Description of dataset attributes and their respective types and scales

Attribute	Type	Subtype	Scale
student_id	Qualitative	Nominal	Nominal
age	Quantitative	Discrete	Ratio
gender	Qualitative	Nominal	Nominal
study_hours_per_day	Quantitative	Continuous	Ratio
social_media_hours	Quantitative	Continuous	Ratio
netflix_hours	Quantitative	Continuous	Ratio
part_time_job	Qualitative	Binomial	Nominal
attendance_percentage	Quantitative	Continuous	Ratio

sleep_hours	Quantitative	Continuous	Ratio
diet_quality	Qualitative	Ordinal	Ordinal
exercise_frequency	Quantitative	Discrete	Ratio
parental_education_level	Qualitative	Ordinal	Ordinal
internet_quality	Qualitative	Ordinal	Ordinal
mental_health_rating	Quantitative	Discrete	Interval
extracurricular_participation	Qualitative	Binomial	Nominal
exam_score	Quantitative	Continuous	Ratio

This classification helps determine appropriate statistical methods for analysis.

4. Data Preprocessing

Data preprocessing was performed to ensure the dataset was clean and suitable for analysis. Missing values were handled by replacing them with appropriate values. Duplicate records were checked and removed to maintain data accuracy. The dataset was then explored using descriptive statistics to understand the overall distribution of variables. Categorical variables were converted into numerical form using label encoding and one-hot encoding. Feature scaling was applied using StandardScaler to normalize the data, and the dataset was split into training and testing sets for model evaluation.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using various visualization techniques including histograms, pie charts, scatter plots, and count plots. These visualizations helped identify patterns and relationships between variables such as study hours, social media usage, and exam scores.

Histograms were used to visualize the distribution of numerical variables such as age, study hours, attendance, and exam scores.

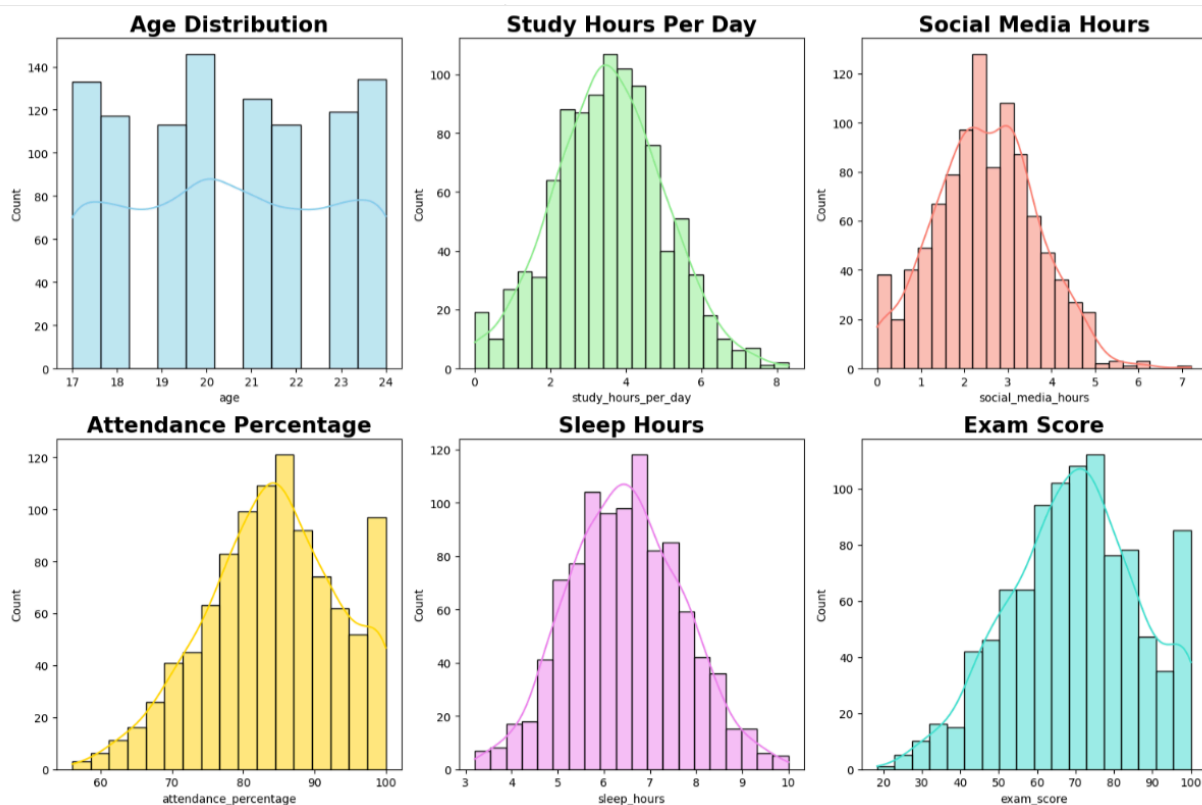


Figure 1: Distribution of key student-related variables including age, daily study hours, social media usage, attendance percentage, sleep duration, and exam scores, illustrating overall trends and variability within the dataset.

Pie charts were used to analyze categorical distributions such as gender.

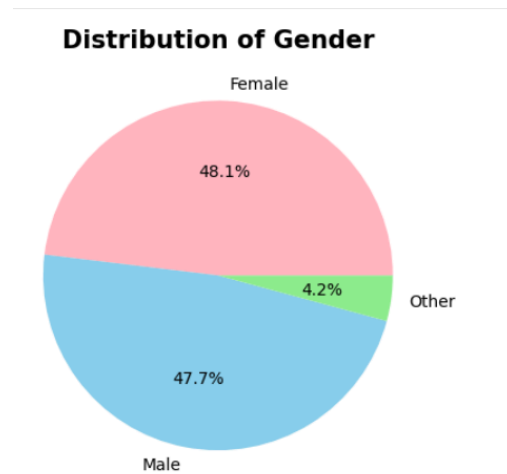


Figure 2: Distribution of gender in the dataset, showing the proportion of male, female, and other participants.

Bar plots are used to compare the distribution or values of categorical variables by representing them as rectangular bars with heights proportional to the corresponding values.

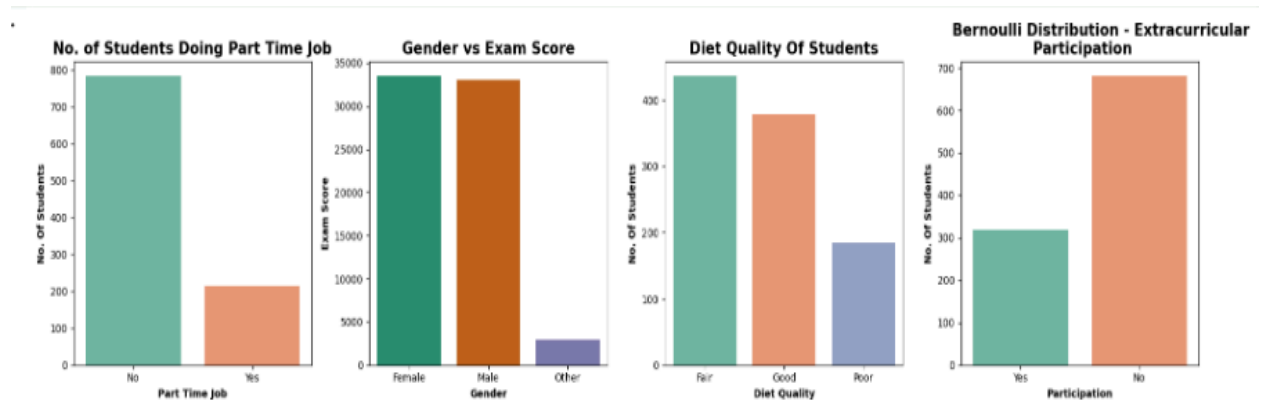


Figure 3: Analysis of student demographics and behaviors, including part-time employment status, gender-wise exam performance, dietary quality, and participation in extracurricular activities.

Scatter plots were used to examine relationships between variables such as study hours and exam scores.



Figure 4: Scatter plots showing relationships between study hours, social media usage, exam scores, and attendance among students.

Correlation heatmaps were used to identify relationships between numerical variables.

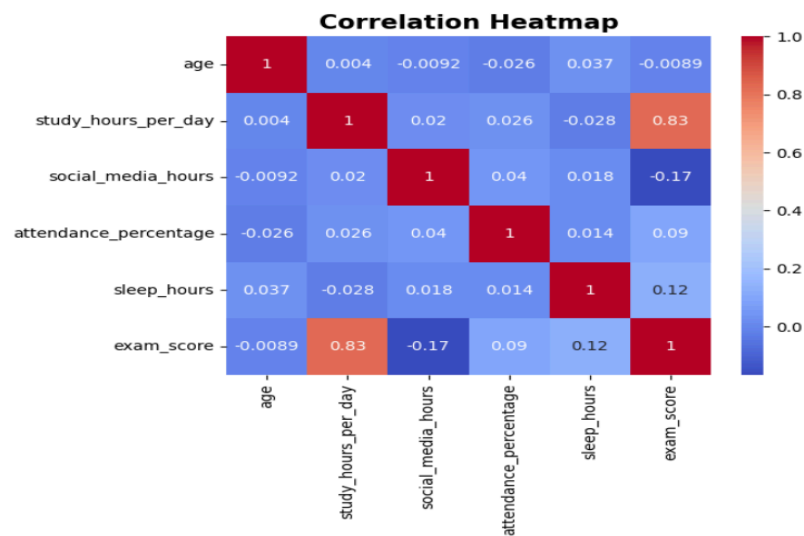


Figure 5: Correlation heatmap showing relationships between student attributes such as study hours, social media usage, attendance, sleep, and exam performance.

6. Machine Learning Model Implementation

After preprocessing and exploratory data analysis, multiple machine learning regression models were implemented to predict student exam scores. The dataset was divided into training and testing sets using an **80:20 ratio**, where 80% of the data was used for training the machine learning models and 20% was used for testing their performance. Each model was trained on the training dataset and evaluated for performance using testing dataset.

- **Linear Regression:** A simple baseline model used to establish a linear relationship between input features and exam scores.
- **Decision Tree Regressor:** A tree-based model that splits the data into branches to capture non-linear relationships.
- **Random Forest Regressor:** An ensemble model that combines multiple decision trees to improve accuracy and reduce overfitting.
- **AdaBoost Regressor:** A boosting technique that combines weak learners sequentially to enhance predictive performance.
- **XGBoost Regressor:** An advanced gradient boosting model known for high efficiency and strong predictive capability.

7. Evaluation Metrics

To assess the performance of the machine learning models, the following evaluation metrics were used:

- **R² Score (Coefficient of Determination):** Measures how well the model explains the variance in the target variable. A higher R² value indicates better model performance.
- **Root Mean Squared Error (RMSE):** Measures the average difference between predicted and actual values. A lower RMSE indicates better accuracy.

These metrics were used to compare different models and identify the most suitable model for predicting student performance.

8. User Interface (Streamlit Application)

A user-friendly interface was developed using Streamlit to allow real-time prediction of student exam scores. The application enables users to input various parameters such as study hours, attendance percentage, sleep hours, social media usage, and other lifestyle factors.

The entered data is preprocessed using the same transformations applied during model training, including encoding and scaling. The processed input is then passed to multiple trained machine learning models to generate predictions.

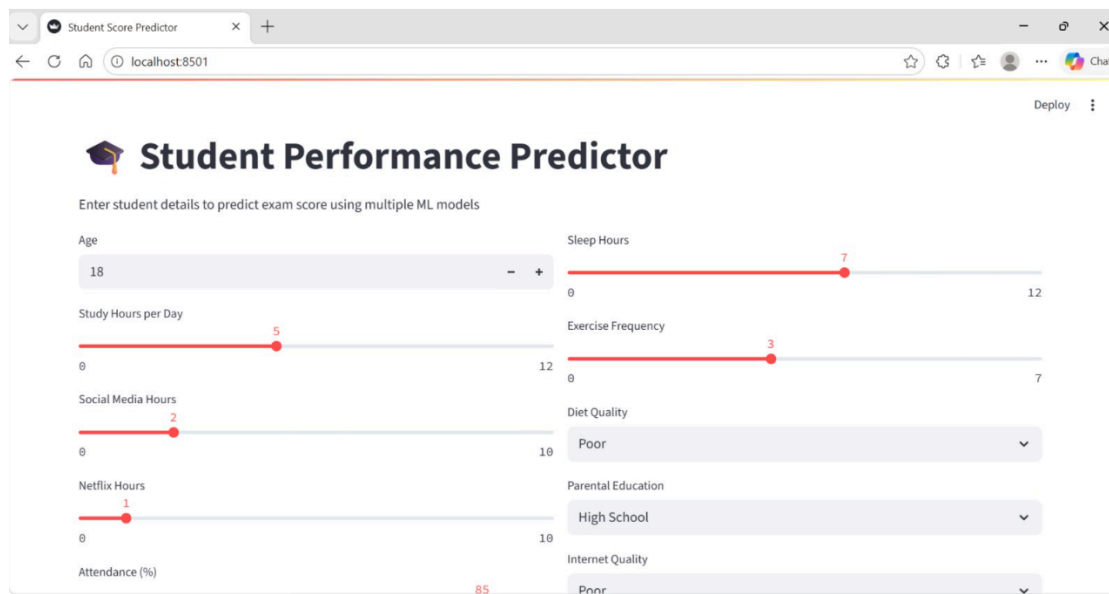
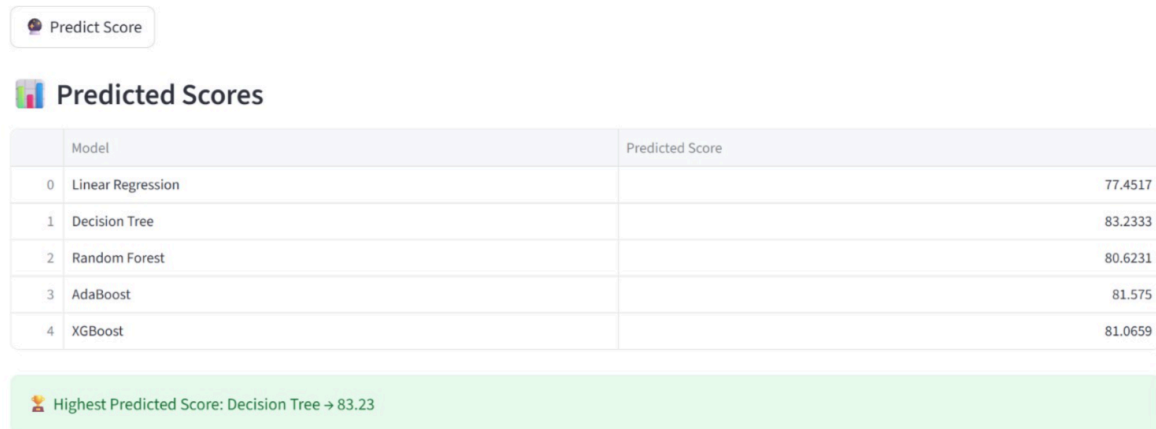


Figure 6: User interface of the Student Performance Predictor

The application displays predicted scores from all models in a tabular format, allowing easy comparison. This interactive system demonstrates the practical implementation of machine learning models and enhances usability for end users.



***Figure 7:** Model comparison showing predicted scores and best model selection.*

9. RESULTS AND ANALYSIS

The exploratory data analysis (EDA) provided meaningful insights into the dataset and helped in understanding the distribution and relationships among variables. Histogram analysis indicated that variables such as study hours, sleep hours, and attendance percentage approximately follow a normal distribution, while exam scores showed noticeable variation among students.

The gender distribution was nearly balanced, ensuring fair representation. Scatter plot analysis revealed a positive relationship between study hours and exam scores, indicating that increased study time contributes to better academic performance. Correlation analysis further supported these findings, showing that study hours and attendance percentage are positively associated with exam scores, while social media usage has a negative impact.

To enhance the analysis, multiple machine learning regression models were implemented to predict student exam scores. The performance of these models was evaluated using R^2 score and Root Mean Squared Error (RMSE).

Model Performance Comparison

Table 2 : Evaluation of different models based on R² Score and RMSE metrics.

Model	R ² Score	RMSE
Linear Regression	0.8988	5.093
XGBoost	0.8770	5.615
AdaBoost	0.8128	6.928
Random Forest	0.8099	6.982
Decision Tree	0.6922	8.884

From the results, **Linear Regression** achieved the highest R² score and lowest RMSE, making it the best-performing model for this dataset. **XGBoost** also demonstrated strong performance, indicating its effectiveness in capturing complex patterns. Ensemble methods such as **AdaBoost** and **Random Forest** showed moderate performance, while the **Decision Tree** model performed the weakest, likely due to overfitting.

Overall, the results indicate that machine learning models can effectively predict student performance, with simpler models like Linear Regression performing surprisingly well for this dataset.

10. CONCLUSION

This study applied data analysis and machine learning techniques to understand and predict student academic performance. The analysis included exploratory data analysis and the implementation of multiple regression models such as Linear Regression, Decision Tree, Random Forest, AdaBoost, and XGBoost.

The exploratory analysis revealed that factors such as study hours and attendance positively influence exam scores, while excessive social media usage negatively impacts performance. These findings highlight the importance of maintaining effective study habits and a balanced lifestyle.

Among the machine learning models, **Linear Regression** achieved the best performance with the highest R^2 score and lowest RMSE, followed by **XGBoost**. Other models such as Random Forest and AdaBoost also produced reasonable results, while Decision Tree showed comparatively lower accuracy.

The study demonstrates that machine learning models can successfully predict student performance based on behavioral and lifestyle factors. It also shows that even simple models can provide strong predictive capability when the data has clear relationships.

Future work can focus on improving model performance by using larger datasets, advanced feature engineering, and hyperparameter tuning. Additionally, deploying the model as a real-time application can further enhance its practical usefulness in educational systems.

11. REFERENCES

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